Parent-Level Debt and Subsidiary-Level Risk-Taking: Evidence from Large Bank Holding Companies *

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Abstract

We investigate the influence of the parent company's capital structure of a bank holding company's (BHC) on the risk-taking behavior of its subsidiary banks, taking into account the BHC's organizational structure. Our analysis shows that increased parent debt is associated with reduced risk-taking by subsidiary banks. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank's risk profile.

Keywords: bank holding company, parent-level debt, bank subsidiary, risk management

JEL Codes: G21, G28, G32, G33, G38

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1 Introduction

Since Jensen and Meckling [1976], corporate finance has a long tradition of considering the incentive implications of capital structure; the study of commercial banks makes no exception. Numerous empirical studies examine the impact of banks' capital structure on their risk-taking incentives, and virtually all studies explaining bank risk-taking would include bank capitalization as a control variable. Yet most modern commercial banks operate within a bank holding company (BHC). Avraham et al. [2012] provide an insightful overview of the organizational structure of US BHCs. Figure 1 reproduces their findings for a typical large BHC, in which the BHC simultaneously controls a bank subsidiary, and a non-bank subsidiary.[1] This multi-tiered organizational structure corresponds to a : the subsidiary bank's own, those of other subsidiaries, the parent company's, and the consolidated balance sheet of the whole group's. This multi-tiered capital structure raises an essential question: which entity's capital position matters? While prior research typically focuses on the capitalization of the commercial bank *or* that of the consolidated BHC, this paper argues that one needs to take a wholistic view, emphasizing the capital position of the parent company. Specifically, we study how the *parent debt ratio*, defined as the ratio of non-deposit debt issued by a BHC's parent company to the BHC's total assets, affects risk-taking by its commercial bank subsidiaries.

The parent debt ratio deserves attention for three reasons. First, parent debt is the typical funding source of subsidiary bank's equity: a parent that borrows and downstreams the proceeds as equity leaves its subsidiary banks well capitalised while the external debt at the BHC level increases. Neither the subsidiary bank's nor the consoli-

[1] Some BHCs also own foreign bank and non-bank subsidiaries.

dated BHC's balance sheet reveals this internal allocation. Second, post-crisis resolution policy has made parent debt the designated loss absorber of the global systemically important banks (GSIBs): under the single point of entry (SPOE) strategy and total loss-absorbing capacity (TLAC) requirements, losses incurred anywhere in the GSIBs are borne by the parent's creditors and shareholders. Third, because parent debt determines whether a subsidiary bank's losses fall on the subsidiary bank's own creditors or are absorbed inside the group, it directly shapes the limited-liability incentives that drive subsidiary bank's risk-taking.

The parent debt ratio shows significant time-series and cross-sectional variation: between 2011 and 2022, the average parent debt ratio rose from 0.72% to 1.50% (Figure 2). The sample in 2021 includes 251 BHCs. 50 of the 251 BHCs had no parent-issued debt, while in the top half of the distribution the average parent debt ratio was 2.63%. Despite such variation, whether parent debt affects the risk-taking of the subsidiary banks that hold most of the group's assets has not, to our knowledge, been examined. We address this gap by examining how the parent debt ratio influences the subsidiary bank' risk-taking—an important question given that commercial banks account for the bulk of BHC assets, rising from 89.03% in 2012Q1 to 94.45% in 2022Q4 (Figure 3). Non-bank subsidiaries enter our analysis only through the model, where their equity determines whether the parent can service its own debt when the subsidiary bank fails.

Theoretically, external debt issued at the parent level may affect the risk-taking of subsidiary banks through two opposing channels. The first is a ***loss-absorption (commitment) channel***. When the parent company raises external debt and downstreams the proceeds to its subsidiary banks as equity investment, the subsidiary bank's capital base expands. The resulting increase in capital constrains the subsidiary banks' incentives to take on risk, since potential losses are more fully absorbed by this equity

buffer. Therefore, this parent-level debt effectively serves as a contractual commitment that helps mitigate risk-taking at the subsidiary level. In contrast, if the parent company issues less debt, subsidiary banks may need to rely more heavily on their own debt issuance to finance investments, which can encourage greater risk-taking due to limited liability. Under limited liability, downside losses are partly borne by debt holders, while shareholders retain the upside gains.

The second is a ***BHC-level risk-shifting channel***. Parent-level debt raises the leverage of the BHC as a whole. Once the parent itself faces a non-trivial probability of default, limited liability at the BHC level shields its shareholders from downside losses, which are transferred to parent creditors. Additional parent debt may therefore amplify, rather than curb, risk-taking incentives across the organization. Moreover, servicing parent-level debt may require the subsidiary bank to upstream dividends, which can push it toward higher-yielding, and hence riskier, assets.

Which channel dominates is ultimately an empirical question. We therefore test these two mechanisms by examining the impact of the parent debt ratio on subsidiary banks' risk-taking. We combine U.S. bank-level regulatory data from the Call Reports with consolidated and parent-only BHC data from the FR Y-9C and FR Y-9LP. We measure subsidiary banks' risk-taking on both sides of the balance sheet. On the asset side, the loan and lease yield captures the ex ante riskiness of new lending, and the provision ratio captures the subsidiary bank's own forward-looking estimate of credit losses. On the liability side, leverage captures the extent to which the bank funds itself with its own debt, which is the channel through which limited liability operates. The Z-score combines both sides into a summary measure of distance to insolvency. Because most subsidiary banks are not publicly traded, these accounting-based measures are the natural choice.

We find that a higher parent debt ratio is associated with lower risk-taking by subsidiary banks. Specifically, an increase in the parent debt ratio is correlated with lower loan yields, lower provision ratios, and higher Z-scores at the subsidiary bank level. These findings suggest that a higher parent debt ratio reduces risk-taking within subsidiary banks.

While the empirical evidence is suggestive, a key concern in identifying the effect of parent debt on subsidiary banks' risk-taking is omitted variable bias. Specifically, when a subsidiary bank faces stringent regulatory capital requirements, it may reduce its risk-taking to avoid potential capital losses. At the same time, the parent BHC may issue additional external debt and inject the proceeds into the subsidiary bank to help it meet these requirements. In this case, the observed negative relationship could be driven by the omitted factor, namely the tightening of regulatory capital constraints, rather than by a causal effect of parent debt ratio.

To address this identification concern, we exploit the introduction of the SPOE resolution strategy and the TLAC requirement in the United States as a natural experiment. These rules unexpectedly exempted BHCs that were subject to the enhanced prudential standards (EPS) of the Dodd–Frank Act but were not classified as global systemically important banks (GSIBs). This exemption allowed large non-GSIB BHCs to sharply reduce their parent-level debt ratios, while the GSIBs' parent-level debt ratios remained elevated because of regulatory constraints. Exploiting this regulatory change, we conduct a difference-in-differences analysis with the large non-GSIB BHCs as the treatment group and BHCs below the EPS threshold as the control group. Importantly, we also use the GSIBs as a placebo treatment group.

Our treatment group, large non-GSIB BHCs, has received little attention in the literature. Empirical studies of resolution reform have focused on the covered firms: Berndt

et al. [2025] estimate market-implied bailout probabilities for U.S. GSIBs, Afonso et al. [2018] compare the ratings and spreads of parent companies and their bank subsidiaries at the four largest BHCs after the SPOE announcement, and Lewrick et al. [2019] study the pricing of bail-in bonds issued by GSIBs and other large banks. Theoretical work likewise centers on the resolution of global banks [Bolton and Oehmke, 2019]. For our question, focusing on the exempted group has two advantages. First, both the GSIBs and the large non-GSIB BHCs were subject to the EPS, so the two groups faced a common set of regulations, with the exception of TLAC, from which the large non-GSIB BHCs were unexpectedly exempted. The exemption therefore changed the expected parent-debt requirement of the large non-GSIB BHCs without altering the other rules they faced. Second, the GSIBs, which share the complexity and supervisory regime of the exempted large non-GSIB BHCs and are closest to them in size, can serve as a comparison group rather than as the object of study.

We find consistent evidence across risk-taking measures that supports the hypothesis that a higher parent debt ratio lowers subsidiary banks' risk-taking. By contrast, we find no comparable change for the GSIB placebo group, which alleviates concerns that our findings are driven by confounding factors, especially those related to BHC size. These results are also consistent with our baseline analysis of nearly 300 U.S. BHCs from 2011 to 2021Q2, which controls for bank- and BHC-level characteristics and includes subsidiary-bank and quarter fixed effects.

We present a simple model that captures our empirical findings. In the model, the BHC comprises both a subsidiary bank and a non-bank subsidiary. The parent company issues debt and allocates the proceeds as equity investments into these subsidiaries. The subsidiary bank faces investment opportunities that involve a trade-off between risk and return. our model predicts that when the parent company issues more debt and invests

the proceeds as equity investment into its subsidiaries, it can mitigate the subsidiary bank's risk-taking behavior. Conversely, if the parent company issues less debt, the subsidiary bank needs to increase its own debt issuance to finance its investments, which can incentivize greater risk-taking. This occurs because, under limited liability, the downside is shared with debt holders, while shareholders pocket the upside.

Our paper highlights the function of internal capital markets within BHCs. Unlike existing studies—some of which focus on risk-sharing through internal capital markets among global banks (e.g., Cetorelli and Goldberg [2012])—we emphasize the role of internal capital markets in shaping the risk-taking incentives of domestic subsidiary banks within U.S. BHCs.

This study contributes to the large literature on the financial ratio regulations aimed at curbing financial institutions' risk-taking incentives. The specific ratio examined here is the parent company's capital structure, which differs from prior research that primarily focus on capital structure at the commercial bank level or the consolidated BHC level (e.g., Keeley [1990], Shrieves and Dahl [1992], Demsetz and Strahan [1997], and Ellul and Yerramilli [2013]).

This paper also contributes to the literature on the organizational structure of financial institutions and risk-taking. Our analysis focuses on the multi-level organizational structures of BHCs and their distinct balance sheets—particularly that of the parent company—rather than the single balance sheet of integrated conglomerates emphasized by Freixas et al. [2007]. Moreover, we highlight the importance of considering existing BHC organizational structures when assessing risk-taking, in contrast to Kahn and Winton [2004] and Luciano and Wihlborg [2018], who design optimal structures to mitigate risk-taking. Closest to our approach is Li et al. [2024], who argue that a bank's

organizational structure should be evaluated alongside its capital structure and bankers' compensation contracts when discouraging banker's risk-taking.

Our paper differs from the SPOE/TLAC literature (e.g., Bolton and Oehmke [2019], Clayton and Schaab [2025]) in two respects. First, that literature studies how parent-level loss-absorbing debt should be designed and deployed to resolve a bank once distress has occurred, whereas we study its ex-ante effect on incentives, before any distress materialises. Second, much of the SPOE literature is set in cross-border banking, where resolution is further complicated because national authorities may ring-fence local subsidiaries and subsidiaries have access to national safety-net subsidies. Our setting, in both the empirical analysis and the model, is purely domestic and abstracts from this additional layer of distortion.

Finally, we add to the nascent literature on parent debt in BHCs, where existing research is scarce. Tucker et al. [2023] show that the parent-only leverage ratio is positively associated with the parent's credit default swap premium, suggesting that parent-only balance sheet data can inform assessments of parent credit risk and that consolidated reporting can obscure critical entity-level details. Building on these insights, we examine how parent company debt affects subsidiary bank risk-taking, offering a novel perspective on risk transmission within BHCs.

The paper is structured as follows: Section 2 describes the data sources, key variables, and summary statistics. Section 3 conducts the empirical analysis on the relationship between parent debt and subsidiary banks' risk-taking. Section 4 presents the difference-in-differences analysis. Section 5 introduces the theoretical model. Finally, Section 6 concludes.

2 Data and summary statistics

In this section, we describe our data sample, outline the construction of the parent debt ratio and subsidiary banks' risk-taking measures, and summarize our sample. Table 1 provides detailed definition of these variables.

2.1 Data sources

2.1.1 BHC-level data

Our BHC-level data are sourced from the Consolidated Financial Statements for Holding Companies (FR-Y9C) and Parent Company Only Financial Statements for Large Holding Companies (FR-Y9LP) provided by the Federal Reserve Bank. We use data spanning from 2011 to 2021Q2, which include quarterly observations of the income statement and balance sheets of all U.S. BHCs and their parent-only companies. We drop BHCs that do not file the FR-Y9C or FR-Y9LP for each quarter during the entire sample period to avoid the impact of BHCs' joining or exiting on our empirical results. Finally, we collect data for 298 BHCs over the sample period.

2.1.2 Subsidiary bank-level data

The subsidiary bank-level data are from the U.S. Call Reports provided by the Federal Reserve Bank, covering the period from 2011 to 2022Q2. This dataset contains quarterly observations of the income statement and balance sheets of all U.S. commercial banks, some of which are the subsidiaries of BHCs. The data contain bank-level identifier

(RSSD9348) that can be used to link to the BHC-level data. Finally, our balanced panel consists of 579 unique subsidiary banks.

2.2 Parent debt ratio

Our key explanatory variable is the parent debt ratio, defined as the parent's liability scaled by the BHC's total consolidated assets. The parent's liability includes the parent company's deposits, securities sold under agreements to repurchase, borrowings of commercial paper and other borrowings with a remaining maturity of one year or less, other borrowed money with a remaining maturity of more than one year, and subordinated notes and debentures.

2.3 Subsidiary bank's risk-taking

We measure subsidiary bank risk along both the asset and liability sides of the balance sheet. Our first measure is the Z-score, defined as the sum of the return on assets (ROA) and the equity-to-assets ratio, scaled by the standard deviation of ROA, where the equity-to-assets ratio is book equity divided by total assets. The Z-score thus captures both asset-side risk, through the level and volatility of returns, and liability-side risk, through capitalization; a higher Z-score indicates greater bank stability. Because the Z-score is highly skewed, we use its natural logarithm, which is approximately normally distributed (Laeven and Levine [2007]). Second, similar with Acharya et al. [2017], we measure leverage as the ratio of the subsidiary bank's debt to its equity. Third, we use two measures that capture risk-taking in the loan portfolio more directly. The loan and lease yield, defined as quarterly interest income on loans and leases divided by quarterly average loans and annualized, is an ex ante measure: higher yields indicate lending to

riskier borrowers. The provision ratio, defined as provisions for credit losses divided by quarterly average loans and leases and annualized, reflects expected credit losses in the loan portfolio (Hirtle et al. [2020]).

2.4 Control variables

We use market data from the Center for Research in Security Prices (CRSP) and regulatory accounting data from Call Reports, FR Y-9C, and FR Y-9LP to construct subsidiary bank- and BHC-level controls that are considered as important determinants of a bank's risk-taking behavior.

These control variables include subsidiary bank size ($LnTA$), the subsidiary bank's leverage ratio ($Debt/Equity$), high-quality liquid assets ($HQLA$), and the BHC's Tobin's Q ($TobinQ$). Keeley [1990] shows that Tobin Q, as a proxy of market power or charter value, is negatively related to a bank's risk-taking. Boyd and Runkle [1993] suggest a negative relationship between bank size and bank risk. Acosta-Smith et al. [2020] find that a leverage ratio requirement associated with the introduction of the Basel III Leverage Ratio can incentivize banks bound by it to increase their risk-taking.

2.5 Summary statistics

Table 2, Panel A summarizes the sample statistics for the 2012-2021Q2 period. For the BHC-level variables, the parent debt ratio averages 1.87%. The average Tobin's Q for BHCs is 1.03. The mean value of HQLA ratio is 15.42%.

For the subsidiary bank-level variables, the mean logarithm of asset size is 15.02, which translates to a mean of USD 3.34 billion in total assets. Regarding subsidiary

banks' risk-taking measures, the mean logarithm of the Z-score is 3.65, the average loan and lease yield is 4.82%, the provision ratio is 0.32%, and the mean leverage ratio is 7.29.

3 Empirical approach and results

The central premise of our paper is the impact of parent debt ratio on the risk-taking of subsidiary banks. In this section, we present evidence that establishes the negative link between the parent debt ratio and subsidiary bank's risk-taking.

For each subsidiary bank, we estimate the following model:

$$\begin{aligned} RiskTaking_{ijt} = \alpha + \beta ParentDebtRatio_{j,t-4} + \\ + \gamma Controls_{i,t-4} + \delta Controls_{j,t-4} + \eta_t + \epsilon_{ijt} \end{aligned} \tag{1}$$

where, for each subsidiary bank i affiliated with BHC j and in quarter t, $ParentDebtRatio$ is calculated as the parent's liability divided by the BHC's total assets, $Controls$ represents a set of subsidiary bank-level and BHC-level control variables described in Section 2.4. The BHC-level control variables are $HQLAratio$ and $TobinQ$. Since most subsidiary banks are not public, we could not calculate the bank-level Tobin Q. Therefore, we use BHC-level Tobin Q as an alternative in our regression model. All explanatory variables are lagged by four quarters to mitigate concerns over look-ahead bias. $RiskTaking$ is a measure of bank's risk-taking originating in quarter t described in Section 2.3. We include subsidiary bank and quarter fixed effects to mitigate omitted-variable bias. In all models, standard errors are clustered at the subsidiary bank level.

Table 3 presents the empirical evidence showing a negative relationship between the parent debt ratio and the subsidiary bank's risk-taking during the 2011–2021Q2 period. Specifically, for a one standard deviation increase in the parent debt ratio (0.033), we observe an expected 9.80% increase in Z-Score (2.97 x 0.033 x 100% = 9.80%), a 0.20% decrease in loan yield (-0.062 x 0.033 x 100% = -0.20%), a 0.16% decrease in provision ratio (-0.048 x 0.033 x 100% = -0.16%), and a 55.08% decrease in leverage (-16.91 x 0.033 x 100% = -55.80%).

To compare magnitudes across outcomes, we scale each of these changes by the standard deviation of the respective outcome variable. The effects are economically meaningful. For the Z-score, whose standard deviation is 0.579, a one-standard-deviation increase in the parent debt ratio (0.033) is associated with an increase of 0.17 standard deviations (2.97 x 0.033 / 0.579 = 0.17). By the same measure, a one-standard-deviation increase in the parent debt ratio is associated with decreases of 0.13 standard deviations in the loan yield (-0.062 x 0.033 / 0.0157 = 0.13), 0.23 standard deviations in the provision ratio (-0.048 x 0.033 / 0.0068 = -0.23), and 0.18 standard deviations in the leverage ratio (-16.91 x 0.033 / 3.0585 = -0.18).

Taken together, these magnitudes indicate that parent-level debt meaningfully reduces subsidiary banks' risk-taking.

4 Institutional Background and Identification

Identifying how a BHC's parent debt affects its subsidiary banks' risk-taking is difficult because the identification of exogenous changes in BHC's debt structure is an empirical challenge. We address this challenge through the introduction of the single point of

entry (SPOE) resolution strategy and the gradual introduction of total loss-absorbing capacity (TLAC) requirement in the United States. The episode is useful for two reasons. First, for the BHCs within their scope, SPOE and TLAC changed how limited liability allocates losses between the creditors of the parent and the creditors of its subsidiaries. Second, which BHCs would fall within that scope was not settled in advance. The Dodd–Frank Act did not limit the new regime to the largest banks, and BHCs with total consolidated assets of $50 billion or more, the threshold for resolution planning and enhanced prudential standards, had reason to expect that it would apply to them. When regulators confined TLAC to the eight U.S. global systemically important banks (GSIBs) in December 2015, large non-GSIB BHCs experienced an unexpected relaxation of anticipated regulation. This group, which studies of resolution reform centered on GSIBs largely overlook, is the focus of our identification strategy.

4.1 Regulatory Background

Before these reforms, the parent and its subsidiaries were resolved as separate legal entities protected by limited liability. A failing bank subsidiary is placed in FDIC receivership on its own; its losses fall on its own creditors and on its shareholder, the parent, and the parent's creditors lose only to the extent that the parent's equity in the subsidiary loses value. Debt issued by the parent and debt issued by the subsidiary bank are therefore exposed to the losses of different legal entities.

SPOE changes this allocation. Under the strategy that the FDIC described in December 2013, only the top-tier holding company enters receivership. Its assets, consisting mainly of its equity in and loans to its subsidiaries, are transferred to a bridge company, whereas its shareholders and unsecured creditors remain in the receivership and absorb

losses. The operating subsidiaries, including the insured depository institution, continue as going concerns and are recapitalized with resources from the parent.[2] Losses incurred anywhere in the group are thus pushed up to the parent's creditors, while the external creditors of the subsidiaries are shielded. Parent-level creditors bear losses that ordinary limited liability would have left with the subsidiary.

SPOE is credible only if the parent holds enough equity and long-term unsecured debt to absorb these losses, and TLAC makes it operable. For each covered firm, the Federal Reserve's TLAC rule requires the top-tier holding company to maintain minimum amounts of external TLAC and of long-term debt issued by the parent.[3] Its "clean holding company" provisions prohibit the parent from issuing short-term debt to third parties, from entering into qualified financial contracts with third parties, and from having liabilities guaranteed by its subsidiaries. For affected BHCs, parent-level debt therefore changed from a financing choice into a regulatory requirement, and the allocation of external debt between the parent and the subsidiaries was no longer determined by the BHC alone. The following table summarizes the regulatory steps; the rest of this section discusses our identification strategy.

[2]The FDIC notes that the parent's debt "would be used to absorb losses, recapitalize the operating subsidiaries, and allow establishment of NewCo" [Federal Deposit Insurance Corporation, 2013, p. 76617].

[3]External TLAC must equal at least the greater of 18 percent of risk-weighted assets (RWA), plus a buffer of 2.5 percent and the firm's GSIB surcharge, and 7.5 percent of total leverage exposure, plus a buffer of 2 percent. Eligible external long-term debt must equal at least the greater of 6 percent of RWA plus the GSIB surcharge and 4.5 percent of total leverage exposure. Eligible debt must be unsecured, "plain vanilla," governed by U.S. law, and have a remaining maturity of at least one year [Board of Governors of the Federal Reserve System, 2017].

Table 1: Timeline of the SPOE and TLAC reforms

Date	Event
2008–2011	G20 calls for effective resolution regimes and creates the FSB; the FSB's *Key Attributes* establish bail-in as a core resolution power [G20, 2008, 2009a,b, Financial Stability Board, 2011, G20, 2011].
July 2010	Dodd–Frank Act: Section 165(d) requires resolution plans from BHCs with $50 billion or more in assets; Title II creates the Orderly Liquidation Authority.
Dec. 2012	FDIC and Bank of England joint paper on resolving GSIFIs from the top [Bank of England and Federal Deposit Insurance Corporation, 2012].
Sept. 2013	G20 asks the FSB to propose loss-absorbing capacity standards for global systemically important institutions [G20, 2013].
Dec. 18, 2013	FDIC notice on the SPOE strategy; does not specify which firms must hold parent-level loss-absorbing debt [Federal Deposit Insurance Corporation, 2013].
Feb. 18, 2014	Enhanced prudential standards (EPS) final rule for BHCs with $50 billion or more; no long-term debt requirement [Board of Governors of the Federal Reserve System, 2014].
Nov. 10, 2014	FSB consultative TLAC term sheet, limited to GSIBs [Financial Stability Board, 2014].
July 20, 2015	GSIB surcharge rule identifies eight U.S. GSIBs [Board of Governors of the Federal Reserve System, 2015].
Oct. 30, 2015	Federal Reserve proposes TLAC and long-term debt requirements for U.S. GSIBs only [Board of Governors of the Federal Reserve System, 2016].
Nov. 2015	FSB final TLAC standard; G20 endorsement at Antalya [Financial Stability Board, 2015, G20, 2015].
Dec. 15, 2016	Final TLAC rule; compliance from January 1, 2019 [Board of Governors of the Federal Reserve System, 2017].
May 2018	EGRRCPA raises the general EPS threshold to $250 billion (after our sample).

4.2 Identification Strategy

The central identification challenge is to isolate exogenous changes in BHCs' parent-level debt. We address this challenge by exploiting the introduction of SPOE and the gradual implementation of TLAC as a natural experiment, treating 2014 and 2015 as a transition period during which uncertainty about the scope of the new regulatory regime was progressively reduced. Three institutional features motivate our identification strategy.

First, SPOE was not designed exclusively for GSIBs, and the Dodd–Frank Act did not settle which firms would have to hold loss-absorbing debt. The Orderly Liquidation Authority can be applied to any financial company whose failure would threaten U.S. financial stability, and all BHCs with total consolidated assets of $50 billion or more were already required to file resolution plans under Section 165(d). Section 165 also directs the Federal Reserve to apply enhanced prudential standards to the same set of BHCs and allows it to tailor those standards across firms; the TLAC rule was later adopted under this authority. Because SPOE imposes losses on the parent while preserving the operating subsidiaries, it can succeed only if the parent holds sufficient loss-absorbing debt and equity. A parent-level debt requirement could therefore plausibly have been imposed on every BHC above $50 billion, and the FDIC's notice of SPOE in December 2013 did not say otherwise. When the notice was published, it was unclear whether large non-GSIB BHCs would also be expected to prepare for SPOE and to issue additional long-term debt at the parent level.

Second, the EPS rule of February 2014, to some extent, reduced this uncertainty. The rule was the Federal Reserve's main implementation of Section 165 for all BHCs with total consolidated assets of $50 billion or more, and it subjected GSIBs and large non-GSIB BHCs to a common set of capital, liquidity, and risk-management standards.

Yet it imposed no requirement related to SPOE, TLAC, or long-term debt. Had the Federal Reserve intended to require loss-absorbing debt from every BHC above $50 billion, the EPS rule would have been a natural vehicle for doing so; its silence on this point was a first signal that such a requirement might be confined to a narrower set of firms. The rule did not, however, eliminate the uncertainty. Section 165 still allowed the Federal Reserve to add standards for any covered company, and the international work on loss-absorbing capacity that the G20 had requested was not yet complete. Whether large non-GSIB BHCs would have to hold loss-absorbing debt at the parent therefore remained an open question after February 2014.

The third feature is that the sequential design of the TLAC framework resolved this uncertainty during 2014Q4 and 2015 by ultimately excluding large non-GSIB BHCs from the scope of both SPOE and TLAC. Because equity and long-term unsecured debt at the parent are necessary for SPOE to operate, the scope of TLAC effectively determined the scope of SPOE. The FSB's consultative term sheet of November 2014 limited the international TLAC standard to GSIBs, although national authorities remained free to extend similar requirements to other firms. The Federal Reserve's identification of the eight U.S. GSIBs in July 2015 and its proposal of October 30, 2015, which applied the TLAC and long-term debt requirements to those eight firms only, then confirmed that large non-GSIB BHCs would be exempt from the U.S. TLAC requirement; the FSB finalized the international standard in November 2015. Large non-GSIB BHCs thus learned that they would not have to hold parent-level loss-absorbing debt, while they remained subject to the same enhanced prudential standards as GSIBs. For these BHCs, the announcement removed an expected regulatory reason to issue debt at the parent without changing the other rules that apply to them. It is this change in expectations that we exploit.

Therefore, we classify BHCs into three groups based on total consolidated assets as of 2014Q1. The treatment group comprises large non-GSIB BHCs with total consolidated assets of $50 billion or more (TreatBHCs), which were expected to be covered but were exempted. The first comparison group comprises BHCs with total consolidated assets below $50 billion, which were never expected to be covered. The second comparison group comprises the eight U.S. GSIBs, which were expected to be covered and remained so. Once the narrower scope became clear, parent-level debt no longer served a regulatory purpose for large non-GSIB BHCs in the treatment group, which should therefore reduce their reliance on it relative to *both* comparison groups.

GSIBs also serve as a placebo treatment group because they are more comparable to large non-GSIB BHCs and are therefore more likely to share similar time-varying trends. Large non-GSIB BHCs differ from smaller BHCs in terms of size, complexity, and supervisory oversight, so a comparison with smaller BHCs alone could capture time-varying shocks specific to large BHCs rather than the effects of SPOE and TLAC. GSIBs share these characteristics and are subject to the same EPS regime as large non-GSIB BHCs, making them more likely to experience similar time-varying shocks. The key time-varying institutional changes that the two groups were expected to share, but that distinguish them from smaller BHCs, are the introduction of SPOE and TLAC. If our results were driven by trends or shocks common to large BHCs rather than by SPOE and TLAC, a comparison between GSIBs and smaller BHCs should yield similar estimates.

Figure 4 and Figure 5 provide evidence consistent with this design. Figure 5 shows that the parent debt ratios of the treatment and control groups move in parallel until 2014, when the SPOE strategy and EPS rule took effect. After 2014, the ratio for the treatment group declines sharply, whereas the ratio for the control group remains stable, with a slight upward trend. Specifically, the parent debt ratio of the treatment group

decreases from 3.5% in 2014Q1 to 3.0% in 2015Q4, whereas that of the control group increases from 0.4% to 0.5%. Figure 5 shows that the parent debt ratio of GSIBs rises sharply after the U.S. TLAC rule was proposed at the end of 2015, from 9.0% in 2015Q4 to 9.6% in 2017Q3.

We first examine whether risk-taking increases more for large non-GSIB BHCs than for BHCs in the control group after the proposal of the U.S. TLAC rule.

4.3 Econometric framework: Difference-in-differences estimator

We estimate a difference-in-differences model using 2012–2013 as the pre-period and 2016–2017 as the post-period. For each BHC, we specify our baseline model as follows:

$$\begin{aligned} RiskTaking_{ijt} = \alpha + \beta\ Post_t + \gamma\ TreatBHC_{ij} \times Post_t \\ + \lambda\ GSIBs_{ij} \times Post_t + \delta\ Controls_{ijt} + \epsilon_{ijt} \end{aligned} \tag{2}$$

where *Post* is a dummy variable equal to 1 for the 2016–2017 post-period and 0 for the 2012–2013 pre-period, and $TreatBHC$ is a dummy variable equal to 1 for large non-GSIB BHCs subject to the Enhanced Prudential Standards but not SPOE strategy or TLAC requirement. $GSIBs$ is an indicator equal to 1 for GSIBs. *Controls* is a set of subsidiary bank- and BHC-level control variables. α_i denotes subsidiary bank fixed effects. The model does not include the stand-alone $TreatBHC$ dummy variable, because it is subsumed by the subsidiary bank-specific fixed effect α_i. *RiskTaking*

denotes one of our measures of subsidiary bank risk-taking: the loan yield, the provision ratio, subsidiary bank leverage, or the Z-score.

We exclude 2014–2015 from the estimation sample, treating it as the transition period during which the scope of SPOE and TLAC was resolved and parent-level debt structures were adjusted. This follows common practice in difference-in-differences designs with a gradually revealed shock, in which observations recorded while the shock is still unfolding are excluded to avoid attenuating the estimated treatment effect.

We include the interaction $GSIBs \times Post$ alongside our coefficient of interest, $TreatBHC \times Post$, for the following reason. It serves as a placebo test of our identifying assumption. GSIBs were subject to SPOE and EPS from the outset, and their TLAC obligation never depended on how the scope of the rule was resolved during the drafting period. If our identification is valid, the subsidiary banks of GSIBs should exhibit no discernible shift in risk-taking over the same window, and λ should be statistically indistinguishable from zero. A significant λ would instead suggest that a confounding factor common to all large BHCs, rather than the resolution of TLAC uncertainty specific to non-GSIB BHCs, drives our results.

To address the downward bias in standard errors that arises in difference-in-differences estimation with serially correlated outcomes, we follow Bertrand et al. [2004] and take the averages of the observations in our sample in the pre-period (2012–2013) and in the post-period (2016–2017). We will conduct a range of robustness checks and falsification tests, including alternative sample periods, to verify our results and rule out alternative explanations. In one such test, we use the transition period (2014–2015) as the post-period to illustrate the gradual decline in the parent debt ratio shown in Figure 3.

Table 2, Panel B, summarizes the sample descriptive statistics of our key variables for the 2012–2013 and 2016–2017 periods. An average subsidiary bank in our natural experiment sample has log total assets of 14.78 and a parent debt ratio of 2.0%. The average Tobin's Q for BHCs is 1.03. The mean value of HQLA ratio is 15.10%. Regarding subsidiary banks' risk-taking measures, the mean logarithm of the Z-score is 3.68, the average loan and lease yield is 4.92%, the provision ratio is 0.36%, and the mean leverage ratio is 7.35. Table 2, Panel C summarizes the collapsed pre- and post-period sample of the treatment group, and Table 5 reports the regression results. Across all risk-taking measures, relative to BHCs below the $50 billion threshold, $50 billion-plus non-GSIB BHCs ($TreatBHC \times Post$) increase risk-taking following the resolution of TLAC uncertainty, over and above the effect of the preceding EPS shock, relative to BHCs below the $50 billion threshold. In contrast, GSIBs ($GSIB \times Post$) exhibit no significant change, which is precisely the pattern predicted by our identification strategy. The coefficient for the treatment effect ($TreatBHC \times Post$) is statistically significant at the 1% level. Model (1) of Table 4 shows that, on average, large non-GSIB BHCs experienced an increase in loan yield that was 0.006 greater than that of BHCs in the control group, whose loan yield declined modestly over the same period. This effect is robust to the inclusion of bank-level controls in Model (2). Similarly, relative to the control group, large non-GSIB BHCs experienced an average increase of 0.003 in the provision ratio (Model (4)) and of 0.791 in the leverage ratio (Model (10)). Model (8) further shows that large non-GSIB BHCs experienced a decline in the Z-score that was, on average, 0.236 greater than that of the control group. Collectively, these results indicate that large non-GSIB BHCs increased their risk-taking following the introduction of the TLAC rule.

To compare magnitudes across outcomes, we scale each treatment effect by the standard deviation of the respective outcome variable. The effects are economically

meaningful. For the Z-score, whose standard deviation is 0.808, the treatment effect corresponds to a decrease of 0.29 standard deviations (-0.236/0.8080 = -0.29). By the same measure, the treatment effect is associated with increases of 0.18 standard deviations in the loan yield (0.0060/0.0333=0.18), 0.29 standard deviations in the provision ratio (0.0030/0.0104=0.29), and 0.24 standard deviations in the leverage ratio (0.7910/3.3340=0.24).

These magnitudes are comparable to those from the baseline panel regressions, in which a one-standard-deviation increase in the parent debt ratio is associated with changes of 0.13 to 0.23 standard deviations across the same outcomes. In the difference-in-differences analysis, the treatment effects range from 0.18 to 0.29 standard deviations, with signs opposite to the panel estimates, as expected given the decline in parent debt among BHCs in the treatment group. The similarity between the two sets of estimates suggests that the panel associations are not driven primarily by the endogeneity of parent debt.

Taken together, the evidence is consistent with a causal effect of parent debt on the risk-taking of subsidiary banks. As TLAC uncertainty was resolved in their favor, BHCs in the treatment group reduced their parent debt and their subsidiaries increased risk-taking. GSIBs, for which the resolution of TLAC scope carried no new information, exhibit no comparable change.

5 A model of parent debt and subsidiary banks' risk-taking

This section develops a theoretical framework to analyze the relationship between parent debt and the risk-taking behavior of subsidiary banks. Detailed derivations and proofs are provided in the Appendix. The model operates over two periods, denoted as $t = 0$ and $t = 1$. We consider a BHC that wholly owns two distinct entities within its organizational structure: a subsidiary bank and a nonbank subsidiary.

5.1 Period $t = 0$

At $t = 0$, the consolidated balance sheet of the BHC comprises one unit of asset, D_c units of debt, and E_c units of equity. In addition to the consolidated balance sheet, the BHC maintains a parent company only balance sheet, which is subject to Federal Reserve oversight. The parent company holds E_p units of equity and issues D_p units of debt. It downstreams these funds to capitalize both the subsidiary bank and nonbank subsidiary proportionally. Specifically, the parent injects $\alpha(E_p + D_p)$ into the nonbank subsidiary and $(1 - \alpha)(E_p + D_p)$ into the subsidiary bank as equity investments, where α and $1 - \alpha$ represent the relative asset sizes of the nonbank subsidiary and subsidiary bank, respectively.

Under this formulation,

$$E_p + D_p \tag{1}$$

represent the uniform equity capital ratio for both subsidiaries, as capitalized by the parent company. While capital allocation within a BHC is typically endogenous, we

treat the subsidiary capital ratios as fixed parameters. This simplifying assumption can be motivated by the presence of binding regulatory capital requirements at both the subsidiary bank and the BHC levels. The nonbank subsidiary's assets, debt, and equity are denoted by α, D_n, and E_n, respectively. Similarly, the subsidiary bank's assets, debt, and equity are denoted by $1-\alpha$, D_b, and E_b, respectively. The balance sheets for the consolidated BHC, the parent company, the nonbank subsidiary, and the subsidiary bank are summarized in Table 2. By construction, E_b and E_n constitute the assets of the parent company (A_b and A_n).

Table 2: Balance Sheets at $t = 0$

Consolidated BHC		Parent Company		Nonbank Subsidiary		Subsidiary Bank	
1	D_c	A_b	E_p	α	E_n	$1-\alpha$	E_b
	E_c	A_n	D_p		D_n		D_b

At $t = 0$, the subsidiary bank encounters an investment opportunity that yields a payoff R_b with probability $P(R_b)$ and a salvage value S with probability $1 - P(R_b)$. To capture a classical risk-return trade-off, we assume the probability of success is a decreasing, linear function of the target payoff:

$$P(R_b) = 1 - \beta R_b, \tag{2}$$

implying that higher returns are inherently less likely. The payoff distribution of the subsidiary bank's investment is therefore:

$$R = \begin{cases} R_b, & \text{with probability } 1 - \beta R_b, \\ S, & \text{with probability } \beta R_b. \end{cases}$$

To ensure that the limited liability friction is economically meaningful, we impose the following condition on the magnitude of the salvage value:

$$S < 1 - E_p. \tag{3}$$

This condition guarantees that, in the absence of capitalization financed by parent debt ($D_p = 0$), the potential loss from the investment is sufficiently severe to render the subsidiary bank insolvent; that is, $S(1-\alpha) < (1-\alpha)(1-E_p)$. If we were to assume the alternative ($S \geq 1 - E_p$), the equity capital endogenously supplied by the BHC would always be abundant enough to absorb the investment losses, preventing subsidiary bank bankruptcy and neutralizing the core risk-shifting incentives.

The nonbank subsidiary generates a return of R_n. For simplicity, we assume the nonbank activities yield a risk-free return, denoted as r_f. This assumption reflects the reality that many nonbank operations—such as advisory services, asset management, and securities underwriting—generate stable fee income that lacks the default risk typical of bank lending portfolios. Normalizing the risk-free rate to 1, and assuming the subsidiary bank does not exercise deposit market power in its deposit rate setting, we establish:

$$r_b = r_f = 1. \tag{4}$$

This provides a neutral benchmark where the subsidiary bank cannot extract rents on the liability side.

5.2 Period $t = 1$

At $t = 1$, the realized balance sheets for the subsidiary bank are determined by the outcome of the subsidiary bank's investment. The balance sheets for the nonbank subsidiary, and the subsidiary bank are summarized in Table 3.

Table 3: Subsidiaries' Balance Sheets at $t = 1$

Nonbank Subsidiary		Subsidiary Bank (Success)		Subsidiary Bank	
$R_n\alpha$	$R_n\alpha - D_n$	$R_b(1-\alpha)$	$R_b(1-\alpha) - D_b$	$S(1-\alpha)$	$\max\{S(1-\alpha) - D_b, 0\}$
	D_n		D_b		D_b

The consolidated BHC shareholder chooses the risk profile R_b to maximize the expected payoff. The objective function is given by:

$$\begin{aligned}\max_{R_b} \quad \mathbb{E}[L] = P(R_b)\left[R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p\right] \\ + (1 - P(R_b))\left[R_n\alpha - D_n + \max\{S(1-\alpha) - D_b, 0\} - D_p\right], \qquad (5)\end{aligned}$$

where the debt obligations of the two subsidiaries are strictly determined by the uniform parent equity allocation:

$$D_b = (1-\alpha)(1 - E_p - D_p), \qquad (6)$$

$$D_n = \alpha(1 - E_p - D_p). \qquad (7)$$

Before analyzing the equilibrium under the limited liability distortion, it is instructive to establish the first-best investment policy. The first-best risk level is determined by maximizing the expected investment payoff of the subsidiary bank's production technology, without regard to the BHC's capital structure or moral hazard. Taking the

first-order derivative of the expected payoff with respect to R_b yields the first-best solution:

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \tag{8}$$

The subsidiary bank fails if and only if its equity value falls below its debt obligation, $S(1-\alpha) < D_b$. Substituting Equation (6), this condition simplifies to:

$$D_p < (1 - E_p) - S \equiv \widehat{D}'_p. \tag{9}$$

Furthermore, when the subsidiary bank fails, the parent/BHC survives if and only if the residual equity value of the nonbank subsidiary is sufficient to cover the parent's independent debt obligation D_p. This survival condition is $R_n\alpha - D_n - D_p \geq 0$, which, applying Equation (7), reduces to:

$$D_p \leq \frac{\alpha[R_n - (1 - E_p)]}{1 - \alpha} \equiv \widehat{D}''_p. \tag{10}$$

Building on these thresholds, the subsequent analysis examines the BHC's optimal risk-taking decision across the two broad cases (1) no parent/BHC bankruptcy when the subsidiary bank fails, and (2) parent/BHC fails. When the parent/BHC fails, the subsidiary bank can fail or not.

5.3 Case 1: No parent/BHC bankruptcy ($D_p \leq \widehat{D}''_p$)

We begin our equilibrium analysis with the parameter space where $D_p \leq \widehat{D}''_p$. In this regime, the BHC possesses sufficient nonbank equity to fully absorb the parent debt obligation, ensuring it survives even if the subsidiary bank fails on its investment.

Lemma 1. *If and only if* $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$*, it holds that* $\widehat{D}'_p < \widehat{D}''_p$*. Excessive risk-taking occurs when* $D_p < \widehat{D}'_p$*, whereas the first-best risk level emerges when* $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$*.*

See Appendix A1: Proof of Lemma 1.

When $D_p < \widehat{D}'_p$, the subsidiary bank fails under distress (realizing salvage value S), but the BHC survives. Limited liability protection at the subsidiary bank level distorts the risk-taking incentives of the BHC shareholder. Mathematically, solving the FOC of the objective function yields the optimal risk level:

$$R_b^* = \frac{1+\beta(1-E_p-D_p)}{2\beta}. \tag{11}$$

Because the default condition ensures $1-E_p-D_p > S$, it strictly follows that $R_b^* > R_b^{FB}$, confirming that excessive risk-taking occurs.

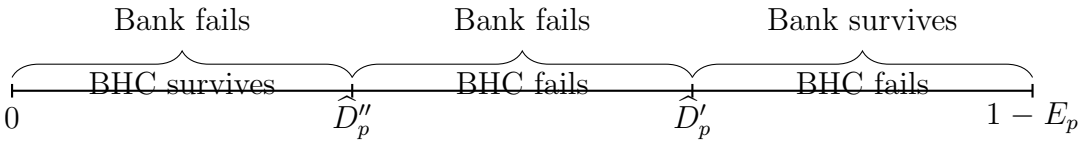
In contrast, when $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$, both the subsidiary bank and the parent/BHC survive the bad state. The losses are fully passed onto the shareholder of the BHC. The absence of limited liability distortion on the subsidiary bank level fully aligns the private risk-taking decision with the first-best level, yielding the first-best outcome, R_b^{FB}.

Lemma 2. *A more significant nonbank subsidiary, i.e., an increase in* α *(given* $\alpha > \widehat{\alpha}$*), reduces risk-taking for a given* D_p*.*

An increase in α acts to increase the parent survival threshold $\widehat{D}''_p$, while the subsidiary bank failure threshold $\widehat{D}'_p$ remains unchanged. This cleanly expands the parameter interval $\left[\widehat{D}'_p, \widehat{D}''_p\right]$ governed by the first-best risk level. Intuitively, when α approaches 0 and the equity value from the nonbank subsidiary becomes negligible, the limited liability distortion is fully transmitted onto the parent/BHC level, creating excessive risk-taking incentives.

Lemma 3. *If and only if* $\alpha < \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$*, it holds that* $\widehat{D}'_p > \widehat{D}''_p$*. Excessive risk-taking occurs when* $D_p < \widehat{D}''_p$ *if the parent/BHC does not fail.*

When α falls below $\widehat{\alpha}$, the ordering of the critical thresholds flips. The possible outcome boundaries are visually summarized in the figure below:



5.4 Case 2: Parent/BHC bankruptcy risk ($D_p > \widehat{D}''_p$)

We now shift to the parameter space where $D_p > \widehat{D}''_p$, exposing the parent company to severe bankruptcy risk. When the parent/BHC fails, its shareholders receive a terminal equity payoff of exactly zero, tightly shielded by parent-level limited liability.

5.4.1 Both the subsidiary bank and the parent/BHC fail ($\widehat{D}''_p < D_p < \widehat{D}'_p$)

Lemma 4. *When both the parent/BHC and subsidiary bank fail, excessive risk-taking still occurs because limited liability distortion emerges at the subsidiary bank level.*

See Appendix A2: Proof of Lemma 4.

Mathematically, the optimal risk level chosen by the BHC shareholder in this regime is given by:

$$R_b = \frac{1 + \beta(1 - E_p - D_p)}{2\beta}. \tag{12}$$

The subsidiary bank still takes excessive risk because the limited liability constraint binds strictly at the subsidiary bank level.

5.4.2 The parent/BHC fails, but the subsidiary bank survives ($D_p > \widehat{D}'_p$)

Lemma 5. *When the parent/BHC fails but the subsidiary bank remains solvent, it yields the first best solution with*

$$R_b^{FB} = \frac{1+\beta S}{2\beta}. \tag{13}$$

See Appendix A3: Proof of Lemma 5.

When the parent/BHC fails but the subsidiary bank remains solvent, the absence of a limited liability distortion at the subsidiary bank level aligns the private risk-taking decision directly with the socially optimal first-best level.

6 Conclusion

This paper provides evidence on how parent debt affects subsidiary banks' risk-taking. Using a comprehensive dataset covering all domestic U.S. commercial banks and their corresponding BHCs from 2011 to 2021Q2, combined with a difference-in-differences analysis using the gradual introduction of SPOE and TLAC as a natural experiment, we find that increased parent debt is associated with reduced risk-taking by subsidiary banks. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank's risk profile.

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Table 1: Variable definitions and sources of data used in the study. This table summarizes the variable definitions and sources of data used in our study.

Variables	Definition
Parent Debt Ratio	Ratio of the parent's non-deposit debt liabilities to the BHC's assets, measured quarterly as: (BHCP2309 (parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332 (parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368 (parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062 (parent company's subordinated notes and debentures) + BHCP0279 (parent company's securities sold under agreements to repurchase)) / BHCK2170 (BHC's total assets)
Loan Lease Yield	Ratio of the subsidiary bank's total interest income from loans to its total loans: (RIAD4010 (total interest and fee income on loans)+ RIAD4065) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or (RIAD4010 (total interest and fee income on loans)+ RIAD4065)/ RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041. Because RIAD4010 and RIAD4065 are reported as year-to-date income, we convert these variables into quarterly amounts before calculating the ratio.
Provision Ratio	Ratio of provisions to total loans: RIAD4230 (provisions for credit losses) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or RIAD4230 (provisions for credit losses) / RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041. Because RIAD4230 is reported as year-to-date income, we convert this variable into quarterly amount before calculating the ratio.
Leverage	Leverage is the subsidiary bank's debt-to-equity ratio: RCON2948 (bank subsidiary's total liabilities, FFIEC 041) / (RCON2170 (bank subsidiary's total assets, FFIEC 041) − RCON2948 (bank subsidiary's total liabilities, FFIEC 041)), or RCFD2948 (bank subsidiary's total liabilities, FFIEC 031) / (RCFD2170 (bank subsidiary's total assets, FFIEC 031) − RCFD2948 (bank subsidiary's total liabilities, FFIEC 031))
Z-Score	Z-Score is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as total equity (RCFD3210, total bank equity capital at the end of the current period) divided by total assets (RCFD2170) for FFIEC 031, or total equity (RCON3210, total bank equity capital at the end of the current period) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Z-Score Narrow	Z-Score Narrow is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as book equity (RCFD3632 (retained earnings) + RCFD3839 (surplus, excluding all surplus related to preferred stock) + RCFD3230 (common stock)) divided by total assets (RCFD2170) for FFIEC 031, or book equity (RCON3632 (retained earnings) + RCON3839 (surplus, excluding all surplus related to preferred stock) + RCON3230 (common stock)) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Ln TA	Total assets is a proxy for subsidiary bank size. It is the logarithm of the subsidiary bank's total assets: ln(RCON2170 (bank subsidiary's total assets, FFIEC 041)), or ln(RCFD2170 (bank subsidiary's total assets, FFIEC 031))

HQLA Ratio	HQLA_ratio is the ratio of the BHC's high-quality liquid assets to the BHC's total assets: (BHCM3531 (U.S. Treasury securities) + BHCM3532 (U.S. government agency obligations, excluding mortgage-backed securities) + BHCKG379 (residential pass-through MBS issued or guaranteed by FNMA, FHLMC, or GNMA) + BHCKG380 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, including CMOs, REMICs, and stripped MBS) + BHCKK197 (commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies) + BHCKG301 (residential pass-through securities guaranteed by GNMA, held to maturity) + BHCKG303 (residential pass-through securities guaranteed by GNMA, available for sale) + BHCKG305 (residential pass-through securities issued by FNMA and FHLMC, held to maturity) + BHCKG307 (residential pass-through securities issued by FNMA and FHLMC, available for sale) + BHCKKX53 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, held to maturity) + BHCKKX55 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, available for sale) + BHCKG313 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity) + BHCKG315 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale) + BHCKK143 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, held to maturity) + BHCKK145 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, available for sale) + BHCKK151 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity) + BHCKK153 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale) + BHCKHT51 (U.S. government agency and sponsored agency obligations, held to maturity) + BHCKHT53 (U.S. government agency and sponsored agency obligations, available for sale) + RCFD0090 (balances due from Federal Reserve Banks) for FFIEC 031, or RCON0090 (balances due from Federal Reserve Banks) for FFIEC 041 + BHCK0213 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, held to maturity) + BHCK1287 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, available for sale)) / BHCK2170 (BHC's total assets)
TobinQ	Tobin's Q is the group-level consolidated Tobin's Q, defined as the ratio of the market value of the BHC (MV) to the replacement cost of its assets (RC). It is estimated as tobinq_ultp = (AT + ME − BE) / AT, where AT is total assets, ME is the market value of equity at year end (stock price multiplied by common shares outstanding), and BE is book equity (stockholders' equity + deferred taxes + investment tax credit − preferred stock). Calculated using the WRDS code; see link for details: https://wrds-www.wharton.upenn.edu/pages/wrds-research/applications/risk-and-valuation-measures/tobins-q-altman-z-score-and-companys-age/

Table 2

Summary statistics of key characteristics. This table summarizes the sample statistics of key subsidiary bank-level and BHC-level characteristics used in our study. Panel A uses the full sample of subsidiary banks and BHCs during the 2011-2021Q2 period. Panel B uses the reduced sample for difference-indifferences models over the 2012-2013 (pre-deregulation) and 2016-2017 (post-deregulation) periods. Panel C uses the collapsed sample of BHCs in the treatment group for difference-indifferences models over the 2012-2013 (pre-deregulation) and 2016-2017 (post-deregulation) periods. The full sample comprise 298 unique BHCs and 579 subsidiary banks.

Panel A: Full Sample (2011-2021Q2)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	N	mean	sd	p10	p25	p50	p75	p90	min	max
Loan Lease Yield	8,907	0.0482	0.0157	0.0372	0.0415	0.0459	0.0509	0.0571	0.0184	0.1364
Provision Ratio	8,907	0.0032	0.0068	-0.0011	0.0002	0.0016	0.0037	0.0087	-0.0099	0.0399
Z-Score	9,871	3.6525	0.5889	3.0928	3.3643	3.6045	3.8726	4.2256	2.0747	6.2843
Z-Score Narrow	9,871	3.6545	0.5785	3.0986	3.3667	3.6063	3.8760	4.2357	2.1379	6.1054
Leverage	9,898	7.2881	3.0621	0.7966	6.3252	7.7454	9.0627	10.3153	0.0076	14.2192
Parent Debt Ratio	9,810	0.0187	0.0328	0.0000	0.0000	0.0084	0.0195	0.0543	0.0000	0.3356
TobinQ	9,898	1.0320	0.0540	0.9795	1.0000	1.0230	1.0531	1.0877	0.9122	1.5686
Ln TA	9,898	15.0257	2.2820	12.5406	13.9228	14.9970	16.3062	17.7974	7.2167	21.8888
HQLA Ratio	9,810	0.1542	0.0811	0.0664	0.0943	0.1391	0.1993	0.2605	0.0025	0.5890

Panel B: Reduced Sample (2012-2013 and 2016-2017)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Loan Lease Yield	4,172	0.0492	0.0160	0.0381	0.0420	0.0466	0.0522	0.0599	0.0184	0.136
Provision Ratio	4,172	0.00361	0.00681	-0.000356	0.000502	0.00192	0.00422	0.00930	-0.00993	0.0399
Z-Score	4,987	3.688	0.631	3.063	3.388	3.638	3.932	4.339	2.075	6.284
Z-Score Narrow	4,628	3.676	0.584	3.103	3.389	3.630	3.907	4.296	2.138	6.105
Leverage	4,643	7.346	3.078	1.459	6.352	7.834	9.078	10.44	0.00762	14.22
Parent Debt Ratio	4,991	0.0199	0.0340	0	0	0.00815	0.0245	0.0608	0	0.336
TobinQ	5,027	1.032	0.0538	0.979	0.998	1.024	1.056	1.089	0.927	1.502
Ln TA	5,026	14.78	2.315	12.28	13.72	14.68	16.04	17.62	0.693	21.49
HQLA Ratio	4,991	0.151	0.0776	0.0653	0.0929	0.138	0.196	0.256	0.00223	0.589

Panel C: Large Non-GSIB BHCs during the Pre-and Post-Period (2012-2013 and 2016-2017)

	2012-2103				2016-2017		
VARIABLES	N	mean	sd	p50	N	mean	sd	p50
HQLA Ratio	36	0.155	0.0440	0.156	36	0.206	0.0472	0.211
Z-Score_post	34	3.676	0.750	3.638	34	3.536	0.881	3.569
Z-Score_nrw_post	26	3.666	0.808	3.653	26	3.606	0.722	3.591
Provision Ratio	20	0.00801	0.0104	0.00324	20	0.00951	0.0150	0.00305
Loan Lease Yield	20	0.0543	0.0333	0.0435	20	0.0533	0.0359	0.0403
Leverage	26	4.984	3.334	6.386	26	5.491	3.559	6.788
TobinQ	35	1.048	0.0913	1.014	35	1.061	0.0700	1.042
Ln TA	36	15.15	3.862	16.17	36	15.33	4.113	16.98
Parent Debt Ratio	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0277
pdratiowda_ultp	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0278
pdratio_ultp	36	0.162	0.113	0.182	36	0.164	0.0774	0.189

Table 3

Parent debt ratio and subsidiary bank's risk-taking. This table presents the results from the model for our preliminary analysis used in Section (4). The dependent variables are the Loan Lease Yield, Provision Ratio, the logarithm of Z-score (Z-Score), and Leverage. The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), (7) and (9) contain only parent debt ratio as a control, while Models (2), (4), (6), (8) and (10) contain all set of controls. All models use quarterly data for 2011-2021Q2. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

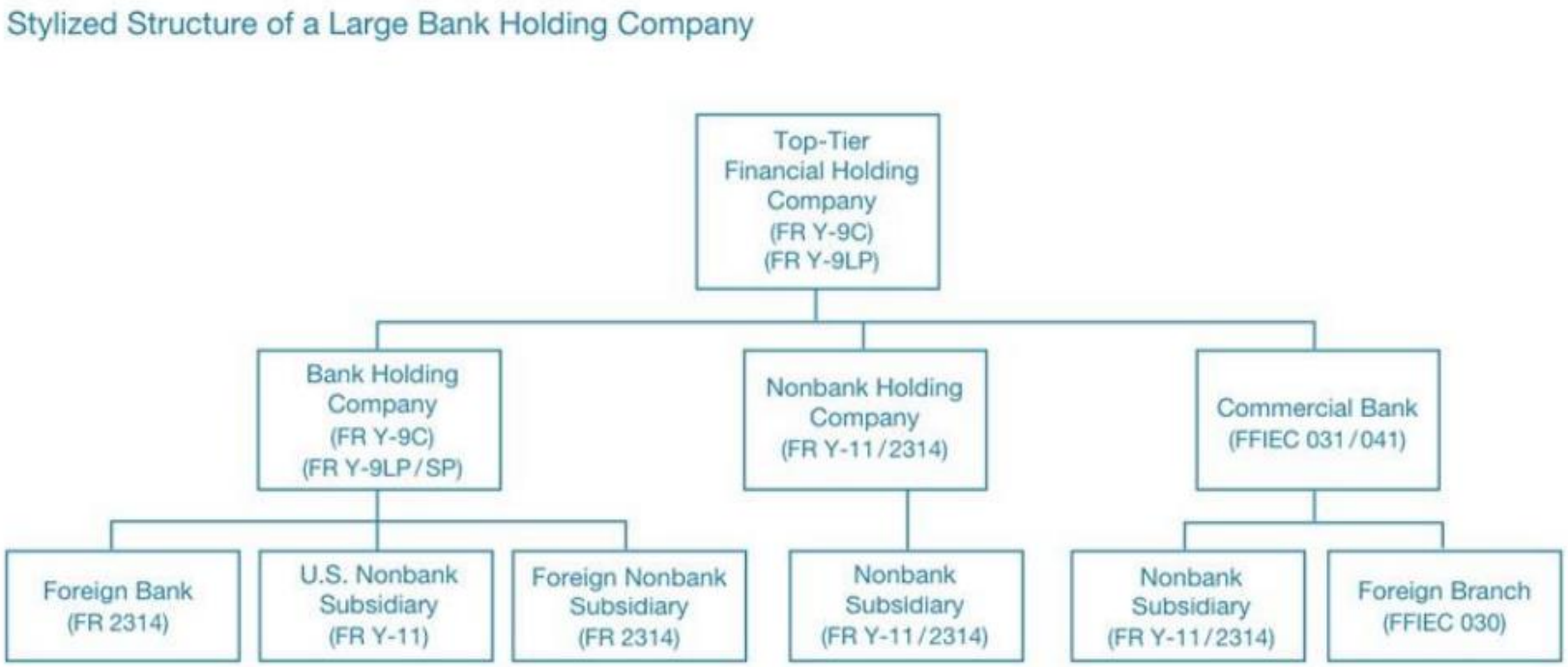
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Loan Lease Yield	Loan Lease Yield	Provision Ratio	Provision Ratio	Z-Score	Z-Score	Z-Score Narrow	Z-Score Narrow	Leverage	Leverage
Parent Debt Ratio	-0.068**	-0.062*	-0.047***	-0.048***	3.966***	2.967***	3.951***	2.973***	-17.295***	-16.905***
	(0.035)	(0.037)	(0.017)	(0.018)	(1.364)	(1.132)	(1.335)	(1.122)	(4.655)	(4.413)
Ln TA		-0.001		0.000		-0.156*		-0.153*		0.041
		(0.001)		(0.000)		(0.092)		(0.089)		(0.272)
Leverage		0.000		0.000		-0.054***		-0.053***		
		(0.000)		(0.000)		(0.013)		(0.012)		
HQLA Ratio		-0.001		-0.008**		0.188		0.147		4.314***
		(0.007)		(0.003)		(0.243)		(0.244)		(0.957)
TobinQ		0.013		-0.004		-0.352		-0.400		-0.392
		(0.010)		(0.004)		(0.336)		(0.336)		(1.190)
Constant	0.049***	0.052***	0.004***	0.002	3.579***	6.662***	3.582***	6.659***	7.608***	6.744
	(0.001)	(0.017)	(0.000)	(0.008)	(0.025)	(1.611)	(0.025)	(1.556)	(0.086)	(4.492)
Observations	8,907	8,907	8,907	8,907	9,871	9,871	9,871	9,871	9,898	9,898
R-squared	0.869	0.870	0.571	0.573	0.655	0.671	0.653	0.669	0.878	0.881
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 4

Difference-in-Differences Analysis in Subsidiary Bank's Risk-taking. This table reports results from difference-in-differences regressions using Equation (2). The treated group includes BHCs with total consolidated assets equal to or greater than 50 billion dollars as of 2014Q1 but excludes GSIBs, while the control group consists of BHCs with total consolidated assets less than 50 billion dollars as of 2014Q1. Post is an indicator variable equal to zero before 2014 and one after 2015. All columns collapse each pre- and post-period into a single observation for each subsidiary bank. Models (1), (3), (5), (7) and (9) contain Nonbank and Time as controls, while Models (2), (4), (6), (8) and (10) contain all set of controls. All models include subsidiary bank fixed effect. T-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Loan Lease Yield	Loan Lease Yield	Provision Ratio	Provision Ratio	Z-Score	Z-Score	Z-Score Narrow	Z-Score Narrow	Leverage	Leverage
Post	-0.007***	-0.008***	-0.002***	-0.002***	0.096***	0.139*	0.098***	0.254***	-0.347***	-0.399**
	(0.000)	(0.001)	(0.000)	(0.001)	(0.031)	(0.080)	(0.031)	(0.072)	(0.075)	(0.168)
TreatBHC x Post	0.006***	0.006***	0.003***	0.003**	-0.225*	-0.244**	-0.139**	-0.236***	0.854***	0.791***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.121)	(0.122)	(0.065)	(0.086)	(0.234)	(0.261)
GSIBs x Post	-0.004	-0.004	-0.001	0.001	-0.138	-0.108	-0.100	-0.181	0.230	0.143
	(0.005)	(0.005)	(0.002)	(0.002)	(0.128)	(0.126)	(0.121)	(0.134)	(0.304)	(0.287)
TobinQ		0.016		-0.029***		-3.013**		-3.243***		2.989
		(0.019)		(0.009)		(1.256)		(1.241)		(2.870)
Ln TA		0.001		0.002***		0.128		-0.108		0.004
		(0.002)		(0.001)		(0.124)		(0.093)		(0.272)
Leverage		-0.000		0.001***						
		(0.001)		(0.000)						
HQLA Ratio		0.008		-0.021***		-0.196		0.050		1.809
		(0.012)		(0.007)		(0.604)		(0.603)		(1.779)
Constant	0.053***	0.022	0.004***	-0.003	3.606***	4.846**	3.602***	8.475***	7.782***	4.395
	(0.000)	(0.046)	(0.000)	(0.017)	(0.015)	(2.298)	(0.014)	(1.998)	(0.035)	(5.005)
Observations	634	634	634	634	728	728	682	682	690	690
R-squared	0.936	0.937	0.792	0.817	0.741	0.750	0.764	0.772	0.947	0.947
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Figure 1: The organizational structure of a large BHC



Data sources: Avraham et al. (2012)

Figure 2: Parent debt ratio over time

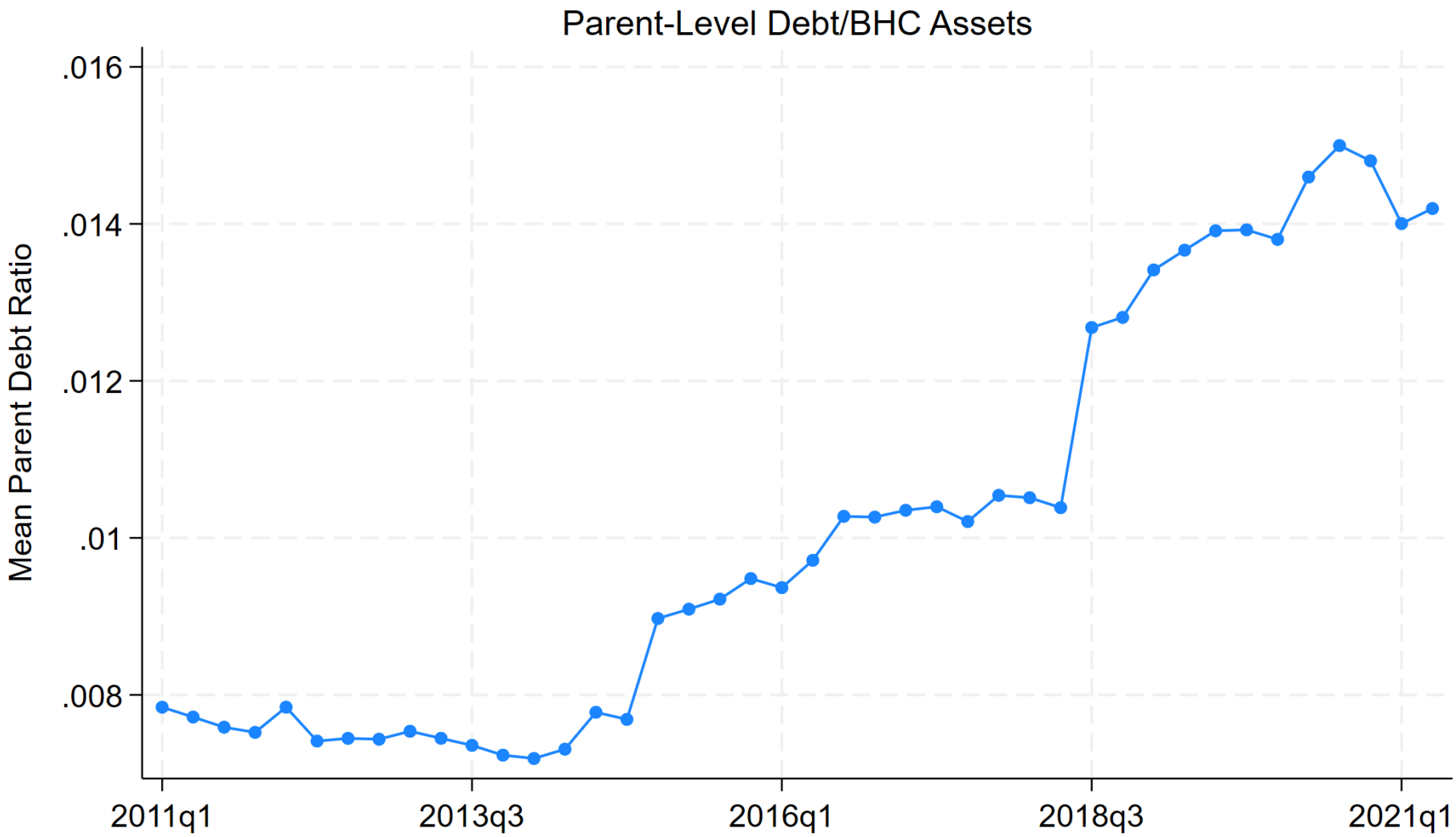


Figure 3: Subsidiary bank's asset ratio over time

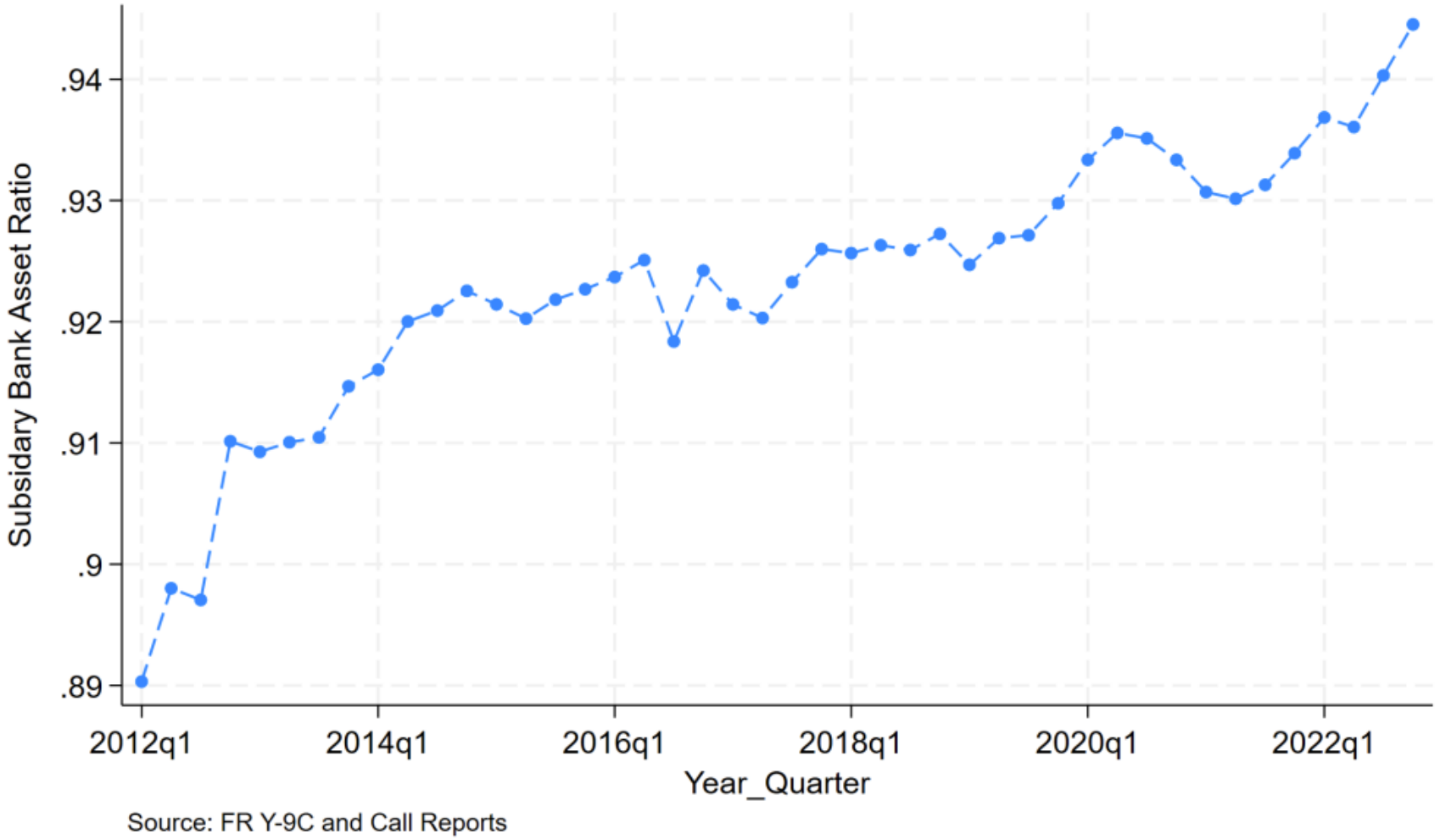


Figure 4: Parallel trend for large non-GSIB BHCs and the control group

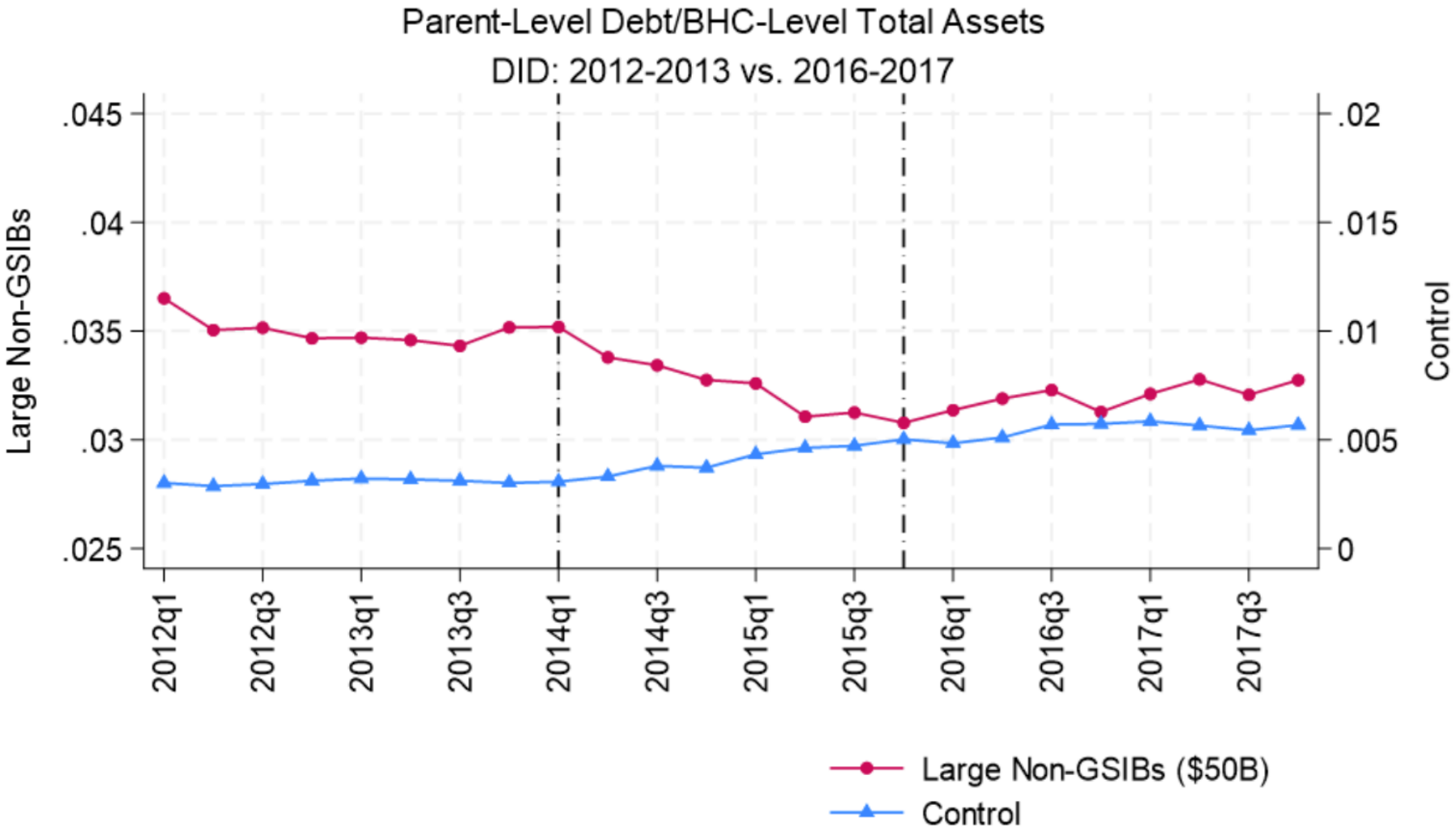
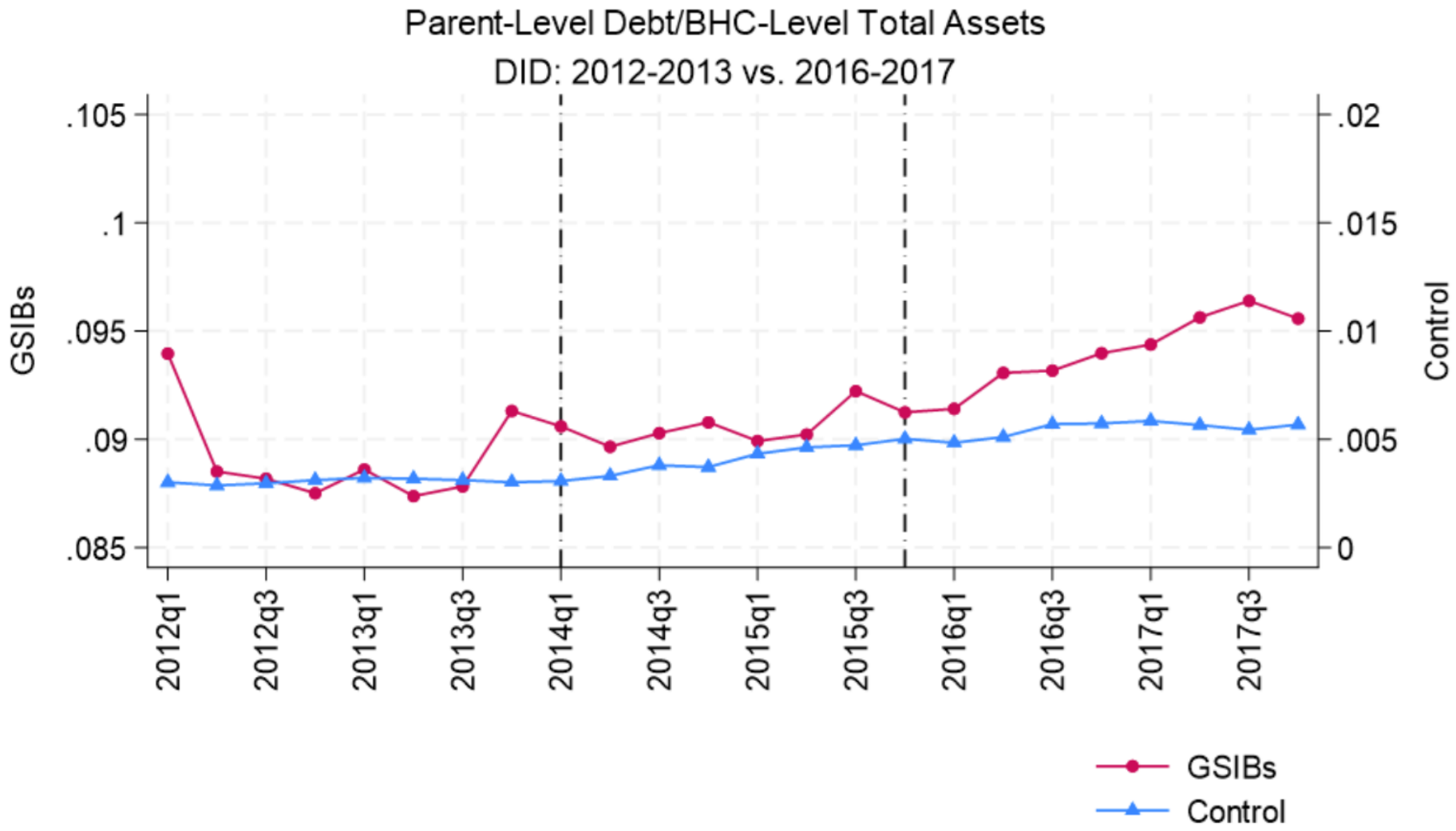


Figure 5: Parallel trend for GSIBs and the control group



Appendix A: Proofs of Lemmas

Appendix A1: Proof of Lemma 1

Lemma 1. *If and only if* $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$*, it holds that* $\widehat{D}'_p < \widehat{D}''_p$*. Excessive risk-taking occurs when* $D_p < \widehat{D}'_p$*, whereas the first-best risk level emerges when* $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$*.*

Proof. Based on Equations (9) and (10), when $S(1-\alpha) - D_b < 0$, the subsidiary bank fails, implying $D_p < \widehat{D}'_p$. The BHC shareholder's payoff is:

$$L = P(R_b)\left[R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p\right] + (1 - P(R_b))\left[R_n\alpha - D_n - D_p\right] \quad (1)$$

Applying Equations (6) and (7) and taking the first-order condition yields:

$$\begin{aligned} &-\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta(1-\alpha)(1-E_p-D_p) = 0 \\ &\Rightarrow 1 + \beta(1-E_p-D_p) = 2\beta R_b \\ &\Rightarrow R_b = \frac{1+\beta(1-E_p-D_p)}{2\beta} \end{aligned}$$

When $S(1-\alpha) - D_b \geq 0$, the subsidiary bank remains solvent, implying $D_p \in \left[\widehat{D}'_p, \widehat{D}''_p\right]$. The BHC shareholder's payoff is:

$$L = P(R_b)\left[R_n\alpha - D_n + R_b(1-\alpha) - D_b - D_p\right] + (1 - P(R_b))\left[R_n\alpha - D_n + S(1-\alpha) - D_b - D_p\right] \quad (2)$$

Taking the first-order condition yields:

$$\begin{aligned} &-\beta R_b(1-\alpha) + (1-\beta R_b)(1-\alpha) + \beta S(1-\alpha) = 0 \\ &\Rightarrow -\beta R_b + (1-\beta R_b) + \beta S = 0 \\ &\Rightarrow R_b = \frac{1+\beta S}{2\beta} \end{aligned}$$

■

Appendix A2: Proof of Lemma 4

Lemma 4. *When both the parent/BHC and the subsidiary bank fail, excessive risk-taking still occurs because the limited liability distortion emerges at the subsidiary bank level.*

Proof. When the parent/BHC fails, market discipline is imposed on parent debt, and the interest rate on parent debt is r_p rather than the risk-free rate. The parent/BHC failure condition requires that the equity value of the nonbank subsidiary is insufficient to cover the parent's debt obligations:

$$R_n\alpha - \alpha(1 - E_p - D_p) - r_pD_p < 0 \tag{3}$$

The subsidiary bank failure condition requires that the subsidiary bank's assets do not cover its debt:

$$S(1 - \alpha) < (1 - \alpha)(1 - E_p - D_p) \tag{4}$$

The creditor's break-even condition is:

$$r_pD_p(1 - \beta R_b) + \beta R_b \left[R_n\alpha - \alpha(1 - E_p - D_p)\right] = D_p \tag{5}$$

The BHC shareholder's payoff is:

$$(1 - \beta R_b) \left[R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p) - r_pD_p\right] + \beta R_b \cdot 0 \tag{6}$$

Substituting the creditor's break-even condition and rearranging yields:

$$(1 - \beta R_b) \left[R_n\alpha + R_b(1 - \alpha) - (1 - E_p - D_p)\right] - D_p + \beta R_b \left[R_n\alpha - \alpha(1 - E_p - D_p)\right]$$

Taking the first-order condition:

$$\begin{aligned}
&- \beta R_b(1 - \alpha) + (1 - \beta R_b)(1 - \alpha) + \beta(1 - \alpha)(1 - E_p - D_p) = 0 \\
&\Rightarrow 1 + \beta(1 - E_p - D_p) = 2\beta R_b \\
&\Rightarrow R_b = \frac{1 + \beta(1 - E_p - D_p)}{2\beta}
\end{aligned}$$

Since the default condition ensures $1 - E_p - D_p > S$, it follows that $R_b > R_b^{FB}$, confirming that excessive risk-taking occurs. ■

Appendix A3: Proof of Lemma 5

Lemma 5. *When the parent/BHC fails but the subsidiary bank remains solvent, the first-best solution obtains:*

$$R_b^{FB} = \frac{1 + \beta S}{2\beta} \tag{7}$$

Proof. The parent/BHC fails only when the subsidiary bank fails, which occurs with probability βR_b. Since market discipline is imposed on parent debt, the interest rate on parent debt is r_p.

The parent/BHC failure condition requires:

$$[R_n\alpha - \alpha(1 - E_p - D_p)] + [S(1-\alpha) - (1-\alpha)(1 - E_p - D_p)] - r_p D_p < 0 \tag{8}$$

The subsidiary bank solvency condition requires:

$$[R_n\alpha - \alpha(1 - E_p - D_p)] + [R_b(1-\alpha) - (1-\alpha)(1 - E_p - D_p)] - r_p D_p > 0 \tag{9}$$

The creditor's break-even condition is:

$$r_p D_p(1 - \beta R_b) + \beta R_b \left[R_n\alpha + S(1-\alpha) - (1 - E_p - D_p)\right] = D_p \tag{10}$$

The BHC shareholder's payoff is:

$$(1 - \beta R_b)\left[R_n\alpha + R_b(1-\alpha) - (1 - E_p - D_p) - r_p D_p\right] + \beta R_b \cdot 0 \tag{11}$$

Substituting the creditor's break-even condition, the payoff simplifies to:

$$\begin{aligned}
&(1 - \beta R_b)\left[R_n\alpha + R_b(1-\alpha) - (1 - E_p - D_p) - \frac{D_p - \beta R_b\left[R_n\alpha + S(1-\alpha) - (1 - E_p - D_p)\right]}{1 - \beta R_b}\right] \\
&= (1 - \beta R_b)\left[R_n\alpha + R_b(1-\alpha) - (1 - E_p - D_p)\right] - D_p + \beta R_b\left[R_n\alpha + S(1-\alpha) - (1 - E_p - D_p)\right] \\
&= R_n\alpha + (1 - \beta R_b)R_b(1-\alpha) + \beta R_b S(1-\alpha) - (1 - E_p - D_p) - D_p
\end{aligned}$$

Taking the first-order condition:

$$\begin{aligned}
&-\beta R_b(1-\alpha) + (1 - \beta R_b)(1-\alpha) + \beta S(1-\alpha) = 0 \\
&\Rightarrow -\beta R_b + (1 - \beta R_b) + \beta S = 0 \\
&\Rightarrow R_b = \frac{1 + \beta S}{2\beta}
\end{aligned}$$

When the parent/BHC fails but the subsidiary bank remains solvent, the absence of a limited liability distortion at the subsidiary bank level aligns the private risk-taking decision directly with the socially optimal first-best level. ■